

Vision Document

Introduction

The project aims to develop a computerized version of the Kivi board game, supporting multiple players and an AI opponent with different difficulty levels.

Problem Statement

The traditional Kivi board game lacks a digital version that allows for flexible gameplay, including human and AI opponents. This project will create an interactive, accessible, and engaging computerized version of Kivi, supporting up to four players, providing different AI difficulty levels, and ensuring accessibility for users with color vision deficiency.

Stakeholders & Key Interests

Stakeholders	Key Interests
Players	Engaging gameplay, multiplayer support, AI opponents
Players with color deficiency	Players with color blindness will also be able to play this game
Game Developers	Creating an accessible and balanced digital game
Educators & Parents	Ensuring the game is suitable for children
Game Publishers	Licensing and Publishing the game. Advertising for the game.

Summary of System Features

- The system shall allow 2 to 4 players to participate in a game.
- The system shall support both human-versus-human and human-versus-computer gameplay.
- The system shall include AI opponents with "Easy" and "Hard" difficulty levels.
- The system shall allow users to save and resume a game session.
- The system shall incorporate accessibility features for players with color vision deficiency.
- The system shall announce the winner at the end of a game
- After the game ends and the winner is displayed the system shall allow the user to start a new game

Project Risks

- Implementing AI with distinct difficulty levels that provide a fair challenge.
- Designing an intuitive and accessible user interface for young players.
- Ensuring color accessibility without compromising gameplay experience.
- Managing game state persistence for the save/load functionality.
- Programming the game in such a way that it can be scaled up to a online networked version